

Event-Driven Multi-Modal Sports Video Understanding: Short Project Report

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1 Problem Statement

Sports video understanding is a challenging task due to dynamic scenes, temporal dependencies, and interactions among multiple entities.

The objective of this project was to develop a deep learning framework capable of detecting important events, modeling tactical interactions, and estimating match intensity for automated sports analysis and future commentary generation.

2 Methodology

A modular pipeline approach was designed because of hardware limitations and large-scale datasets.

Major modules include:

- Feature Extraction
- Event Detection using LSTM
- Tactical Analysis using Graph Neural Networks
- Match Tension Estimation
- Commentary Generation Head

To reduce computational cost:

- Videos were segmented
- Features were extracted offline
- Feature caching was applied

3 Results

Training performance showed gradual learning improvement.

Epoch	Training Loss	Validation Loss
1	1.972	1.854
20	1.246	1.440
40	0.887	1.579

Validation loss reached its minimum around epoch 20 before overfitting began.

The modular architecture successfully improved event recognition performance and enabled scalable training.

4 Conclusion

The proposed framework demonstrates effective multi-modal sports video understanding and establishes a scalable foundation for future automated sports commentary systems.